Exercise 3: Builder Pattern

Aim

To implement the Builder Design Pattern in Java for creating a Computer object with different configurations.

Description

In this exercise, a Computer class is created with attributes such as CPU, RAM, Storage, and Graphics Card. A static nested Builder class is used to set these values and build the object step by step. This approach makes object creation simple and easy to understand.

Classes

Computer

Stores the details of a computer:

- CPU
- RAM
- Storage
- Graphics Card

Builder

Provides methods to set each attribute and returns the completed Computer object.

Builder Pattern Test

Contains the main() method to create and display different computer configurations.

Output

Computer Configuration

CPU: Intel i7

RAM: 16 GB

Storage: 512 GB SSD

Graphics Card: NVIDIA RTX 4060

Computer Configuration

CPU: AMD Ryzen 5

RAM: 8 GB

Storage: 1 TB HDD

Graphics Card: Integrated

Conclusion

This exercise helped me understand how the Builder Pattern can be used to create objects with multiple optional attributes in a clear and organized way. It also improves code readability and avoids using multiple constructors.